



ASSIGNMENT REPORT

on

“EMPLOYEE AI”

(Document Q&A Assistant Using RAG)

Submitted by

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Employee AI – Workplace Intelligence Assistant

1. Introduction

Employee AI is an AI-based workplace assistant developed to help users quickly find information from company policy documents and employee data.

The system provides a chatbot interface where users can ask questions in natural language. It can answer questions related to company policies, employee statistics, and questions that require both types of information.

The project uses Retrieval-Augmented Generation (RAG) for policy documents and Pandas for employee data analysis.

2. Problem Statement

Companies maintain important information in different documents and employee records. Finding the required information manually can take a lot of time.

The aim of this project is to develop an AI assistant that can:

- Search company policy documents.
- Answer employee-related questions.
- Provide reliable numerical information.
- Show the source of policy information.
- Handle questions involving both policies and employee data.

3. Objectives

The main objectives of the project are:

- To develop an easy-to-use workplace chatbot.
- To use AI for searching company policy documents.
- To answer employee data questions accurately.
- To provide document and page references for policy answers.
- To support combined policy and employee questions.
- To reduce incorrect or unsupported AI answers.

4. Technologies Used

Technology	Purpose
Python	Main programming language
Streamlit	Web interface
Ollama	Runs the local AI model

Technology	Purpose
Llama 3.2 3B	Generates answers
FAISS	Stores and searches document vectors
Sentence Transformers	Creates text embeddings
LangChain	Document processing and chunking
Pandas	Employee data analysis
PyPDF	Extracts text from PDF files

5. Dataset

The project uses an employee CSV dataset containing 4,653 employee records across 9 columns.

#	Column	Description
1	Education	Highest qualification of the employee
2	JoiningYear	Year the employee joined the company
3	City	Work location
4	PaymentTier	Salary tier / band
5	Age	Age of the employee
6	Gender	Gender of the employee
7	EverBenchd	Whether ever benched without a project
8	ExperienceInCurrentDomain	Years of experience in current domain
9	LeaveOrNot	Whether the employee left the company

The dataset is used to answer questions such as:

- How many employees are in Bangalore?
- How many employees are in Pune?
- How many female employees are there?
- What is the average experience?
- How many employees joined in 2017?

6. Policy Documents

The system also uses company policy PDF documents as its knowledge base. Some of the documents used are:

#	Policy Document
1	Employee Manual
2	Overtime Policy
3	IT Acceptable Use Policy
4	Credit Card Policy
5	Cost Monitoring and Control Policy

These documents provide information about company rules, procedures, security, overtime, expenses, and other workplace policies.

7. Methodology

The project works in the following steps:

Step	Description
1. Upload Documents	Company policy PDFs are placed inside the documents folder.
2. Extract Text	The system reads the PDF documents using PyPDF.
3. Split Documents	Large documents are divided into smaller sections called chunks.
4. Create Embeddings	Text chunks are converted into numerical vectors using the all-MiniLM-L6-v2 embedding model.
5. Store in FAISS	Vectors are stored in FAISS so relevant information can be searched quickly.
6. User Asks a Question	The user enters a question through the Streamlit chatbot.
7. Query Routing	The system identifies whether the question is Employee-related, Policy-related, or Hybrid.
8. Generate Answer	Policy questions retrieve document sections and use Llama 3.2 via Ollama; employee questions are computed directly with Pandas.
9. Display Result	The answer is shown in the chatbot along with document and page information when applicable.

8. System Architecture

The basic system flow is:

- User → Streamlit Interface → Query Router → Processing → Answer

For policy questions:

- Question → FAISS Search → Relevant PDF Content → Llama 3.2 → Answer + Sources

For employee questions:

- Question → Employee CSV → Pandas → Result

For hybrid questions:

- Question → Policy Processing + Employee Processing → Combined Answer

9. Employee Query Results

The system was tested with several employee-related questions. These results are calculated directly from the employee dataset.

Question	Result
How many employees are in Bangalore?	2,228
How many employees are in Pune?	1,268
How many female employees are there?	1,875
How many male employees are there?	2,778
What is the average experience?	2.9 years
How many employees joined in 2017?	1,108

10. Policy Query Results

The system was tested with policy questions such as:

- What are the overtime rules?
- What does the IT Acceptable Use Policy cover?
- What are the company credit card rules?
- What does the cost monitoring policy require?
- What benefits are available?

The system retrieves relevant sections from the policy documents and generates an answer based on the retrieved information. The response can also show the document name and page number, allowing the user to verify the information.

Sample answer — “What are the overtime rules?”

#	Overtime Rule
1	Exempt employees are not eligible for overtime under any circumstances.
2	Non-exempt employees are not eligible for overtime unless pre-approved in writing by their supervisor and, if applicable, the grants officer.
3	Overtime is compensated at one-and-one-half (1½) times the normal hourly rate for all time worked over 40 hours in a work week.
4	The work week is seven (7) consecutive days, from Monday 12:00 am to the following Sunday 12:00 am.
5	Non-exempt employees must clock in/out daily, and must clock out for breaks of 30 minutes or more.

#	Overtime Rule
6	Non-exempt employees are not eligible for Comp Time, but may be eligible for overtime if authorised.
7	Non-exempt employees must monitor their hours to avoid crossing the 40-hour threshold.
8	Non-exempt program staff may be eligible for overtime if pre-approved by both supervisor and grants officer.

11. Hybrid Queries

The system can answer questions that require both policy information and employee data. For example:

Question: “What are the overtime rules and how many employees are in Bangalore?”

The system provides:

- The overtime policy information from the PDF.
- The Bangalore employee count from the CSV.

This makes the system more useful because it can combine different types of information in one response.

12. Hallucination Control

One important feature of the project is reducing unsupported answers. For policy questions, the AI is instructed to use the retrieved document information rather than making up information. For example:

Question: “Does the company provide pet insurance?”

If the available documents do not contain information about pet insurance, the system indicates that there is no direct information available instead of creating a policy.

13. User Interface

The application is developed using Streamlit and contains the following sections:

Section	Description
Assistant	Users can ask questions and receive answers.
Employee Directory	Users can view employee information and basic statistics.
Documents	Users can see the available policy documents.
System Status	Users can check the status of the CSV data, documents, and vector database.

The interface uses a simple purple-themed design.

14. Evaluation

The system was tested using 15 questions. The evaluation included:

- Policy questions
- Employee questions
- Hybrid questions
- Negative / not-found questions

The testing showed that the system could retrieve policy information, calculate employee statistics, and combine both types of information.

One question related to average employee age should be checked again during the final test before submitting the report.

15. Limitations

- The application currently runs locally.
- It is not yet connected to Microsoft Dataverse.
- PDF extraction may not work perfectly for complex documents.
- The local Llama 3.2 3B model is smaller than large cloud-based AI models.
- Employee queries currently support predefined question patterns.

16. Future Scope

- Connecting employee data to Microsoft Dataverse.
- Integrating the system with Power Apps.
- Adding user authentication.
- Adding role-based access.
- Allowing users to upload new documents.
- Adding more employee analytics.
- Deploying the application for enterprise use.

17. Screenshots

The screenshots below show the working Employee AI application.

Screenshot 1 – Employee AI Assistant (Chat Interface)

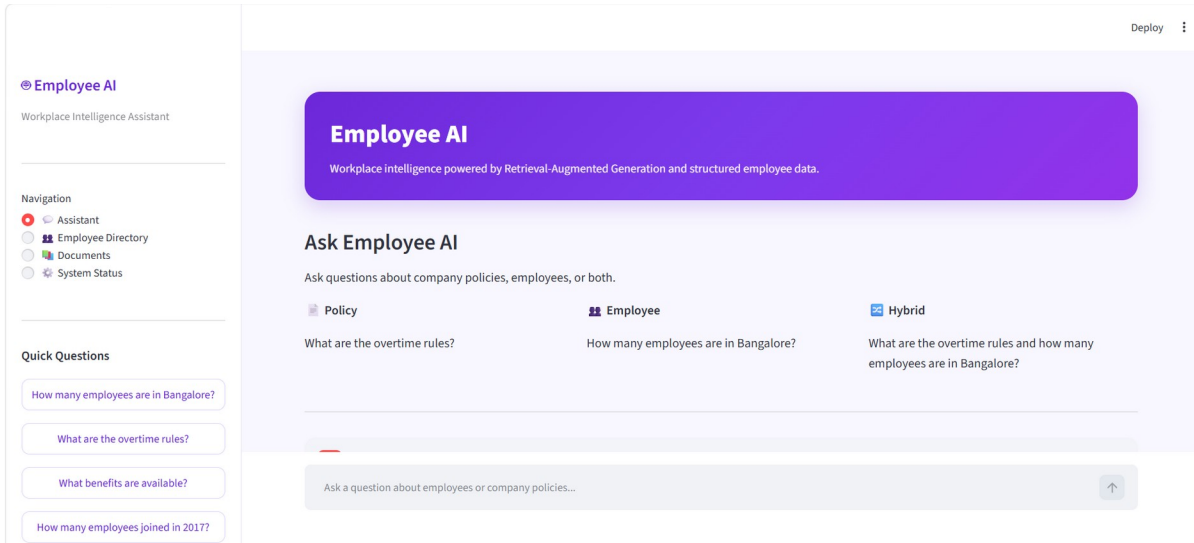


Figure 1: The Assistant tab, showing Policy, Employee, and Hybrid quick-question shortcuts.

Screenshot 2 – Policy Question and Sources

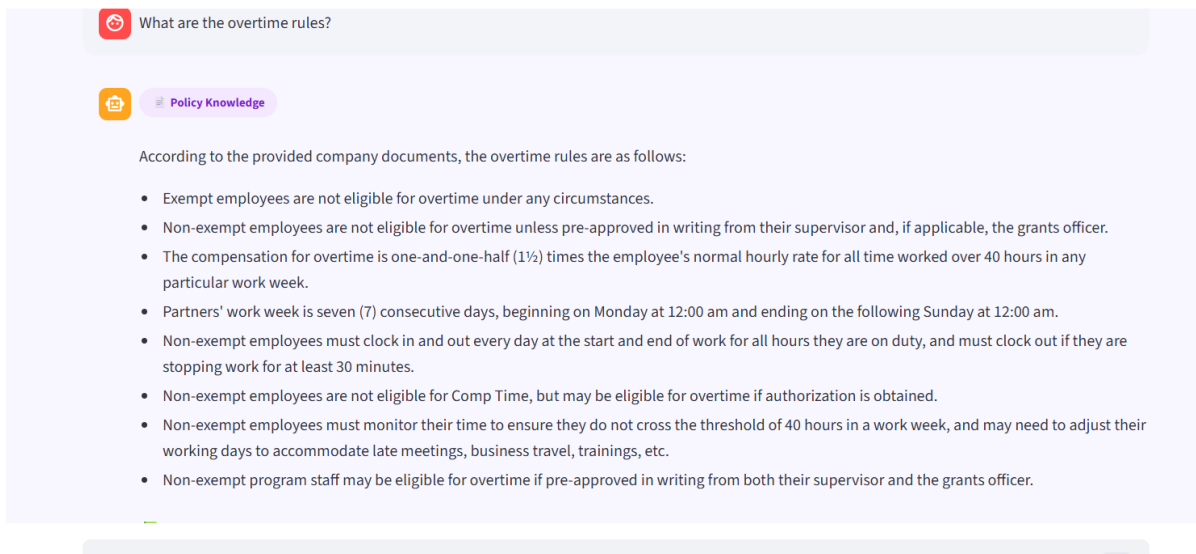


Figure 2: Response to “What are the overtime rules?”, tagged as Policy Knowledge.

Screenshot 3 – Employee Directory

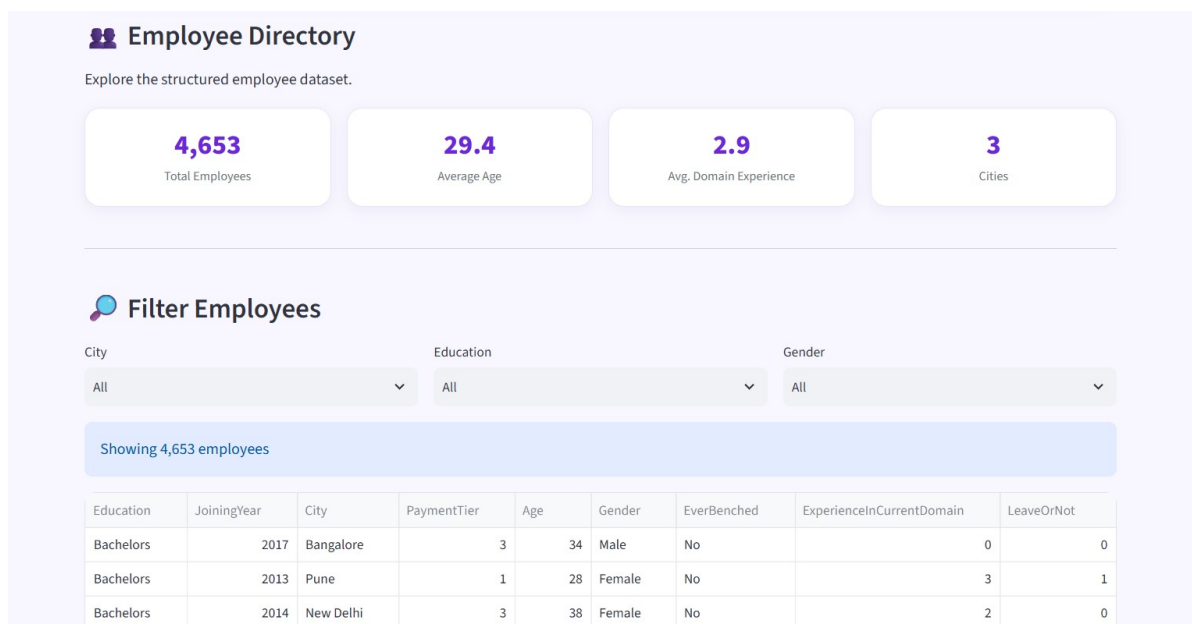


Figure 3: Employee Directory with summary metrics and filterable employee table.

Screenshot 4 – System Status

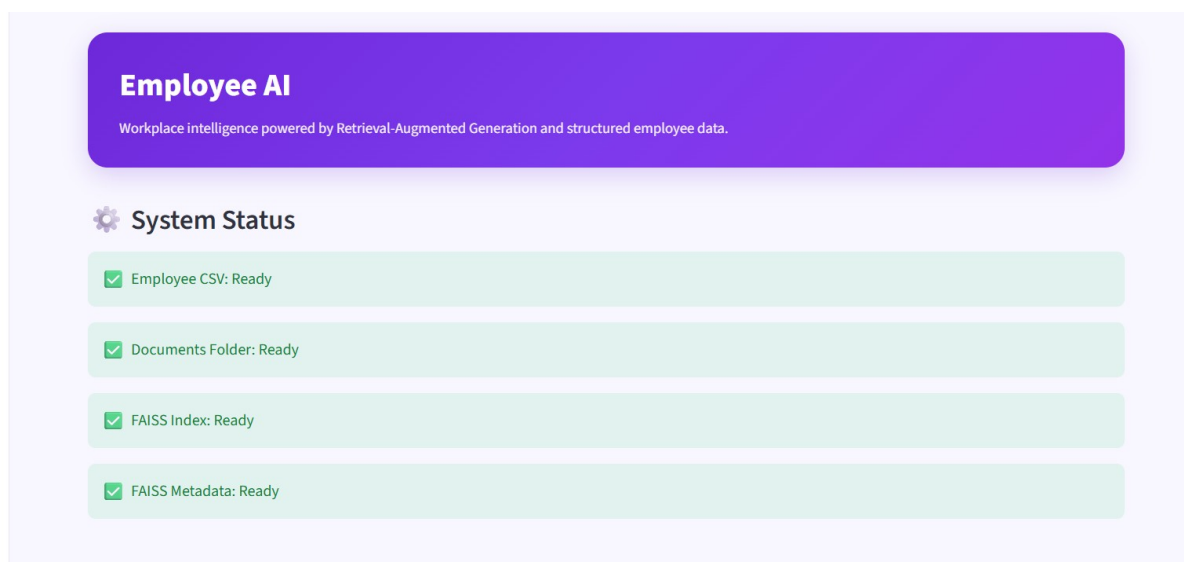


Figure 4: System Status panel confirming the CSV, documents folder, FAISS index, and FAISS metadata are all ready.

18. Conclusion

Employee AI is a workplace intelligence assistant that combines AI-based document search with employee data analysis.

The system can answer questions from company policy documents, calculate employee statistics, provide source information, and handle hybrid questions.

The project demonstrates how RAG, LLMs, vector search, and structured data analysis can be combined to create a useful workplace assistant.

The current prototype can later be extended with Microsoft Dataverse and Power Apps for enterprise deployment.